

Some Main Ideas in the Work of A. Grothendieck

Pierre Deligne*

Abstract. This article attempts to explain four fundamental mathematical concepts created by Grothendieck: schemes, topoi, the six operations, and motives.

In *Recoltes et semailles* (part three), Grothendieck writes:

“Take for example the task of proving a theorem that remains hypothetical, to which, for some people, mathematical work would seem to reduce. I see two extreme approaches to going about it. One is that of the hammer and chisel, when the problem posed is seen as a large nut, hard and smooth, whose interior one must reach, the nourishing flesh protected by the shell. The principle is simple: one puts the edge of the chisel against the shell and strikes hard. If necessary, one repeats this at several different places, until the shell breaks—and one is pleased. [...]

“I could illustrate the second approach, still keeping the image of the nut that must be opened. The first parable that came to my mind just now is that one plunges the nut into a softening liquid, simply water, why not; from time to time one rubs it so that it penetrates better, and for the rest one lets time do its work. The shell softens over weeks and months—when the time is ripe, pressure from the hand suffices, and the shell opens like that of an avocado ripened to the point. Or else one leaves the nut to ripen under the sun and the rain, and perhaps also under the frosts of winter. When the time is ripe, it is a delicate shoot emerging from the substantial flesh that will have pierced the shell, as if in play—or better said, the shell will have opened by itself, to let it pass. [...]

“The reader who is at all familiar with some of my work will have no difficulty recognizing which of these two modes of approach is ‘mine.’” [1985, p. 552–553]

A little further on [ibid., p. 554, note (***)], Grothendieck puts forward four examples: Riemann–Roch, the prime-to-the-characteristic structure of π_1 for curves, the rationality of L -functions for schemes of finite type over a finite field, and the semistable reduction theorem for abelian varieties.

I remember my astonishment in 1965–66 after Grothendieck’s lecture [SGA 5] proving the base-change theorem for $Rf_!$: devissages, more devissages, nothing seems to happen, and yet at the end of the lecture a clearly nontrivial theorem is there.

Many of Grothendieck’s ideas have become so familiar to us, and are so perfectly suited to their object, that we forget that they were far from obvious at birth; we even forget their author. My purpose in this article is to describe four of these ideas: schemes, topoi, the six operations, and motives.

1 Schemes

The invention of schemes is the first of Grothendieck’s ideas that one thinks of, perhaps because it was the one accepted most quickly. Serre’s lecture in Stockholm in 1962 begins: “I would like

*Institute for Advanced Study, School of Mathematics, Princeton, NJ 08540, USA.

to present here some of the recent developments in algebraic geometry. I should specify that I take this last term in the sense it has acquired over the last few years: that of the theory of schemes.” This acceptance was facilitated by the rapid publication, thanks to Dieudonné’s collaboration, of EGA.

The boldness of Grothendieck’s definition is to accept that every commutative ring with unit A defines an affine scheme $\text{Spec}(A)$; that is, not to try to restrict oneself to a category of “good” rings: integral, reduced, noetherian, and so on. This has a price. The points of $\text{Spec}(A)$, namely the prime ideals of A , do not have a manageable geometric meaning, and the structure sheaf $\mathcal{O}$ is not a sheaf of functions. When one has to construct a scheme, one generally does not begin by constructing its set of points.

Perhaps more important is the decision to build a relative theory, witnessed by the omnipresent diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ S' & \longrightarrow & S \end{array}$$

in Grothendieck’s lectures. The classical case of a variety defined over a field k becomes the special case $S = \text{Spec}(k)$. In a relative theory, with an S -scheme X , i.e. a scheme X over S , or equivalently with a morphism of schemes $f : X \rightarrow S$, one systematically considers the scheme X' deduced from X by a base change $u : S' \rightarrow S$, that is, the fiber product $X' := S' \times_S X$ and its projection f' to S' :

$$\begin{array}{ccc} X' & \longrightarrow & X \\ f' \downarrow & & \downarrow f \\ S' & \xrightarrow{u} & S \end{array}$$

In the category of schemes, fiber products always exist: if allowing every commutative ring to define an affine scheme gives bizarre schemes a right to exist, it also provides a category of schemes with good properties.

A property of X over S will be called *geometric* if it has good invariance properties under base change. Classical analogue: for a variety X defined over a field k , the set of points of X over an algebraically closed extension $\bar{k}$ of k (for example Weil’s universal domain) is considered “geometric,” while the set of k -points is “arithmetic.”

If X is a scheme over S , and $u : S' \rightarrow S$ is a morphism of schemes, an S' -point p of X is a morphism of S -schemes:

$$\begin{array}{ccc} & & X \\ & \nearrow p & \downarrow \\ S' & \longrightarrow & S \end{array}$$

We write $X(S')$ for the set of S' -points of X . It identifies with the set of sections of $X' \rightarrow S'$.

Example 1. Let X be the affine scheme over a field k defined by equations $P_\alpha(X_1, \dots, X_n) = 0$:

$$X = \text{Spec}(k[X_1, \dots, X_n]/(P_\alpha)).$$

Let k' be a k -algebra. The set $X(k') := X(\text{Spec}(k'))$ is the set of solutions in k'^n of the equations $P_\alpha = 0$. For k' an algebraically closed extension $\bar{k}$ of k , this is the set $X(\bar{k})$ that Weil views as underlying X .

Example 2. The scheme GL_N over $\text{Spec}(\mathbb{Z})$ is such that, for every commutative ring A —automatically a $\mathbb{Z}$ -algebra, so $\text{Spec}(A)$ is a scheme over $\text{Spec}(\mathbb{Z})$ — $\text{GL}_N(\text{Spec}(A))$ “is” $\text{GL}_N(A)$.

In these examples, the geometric intuition one has of X , a scheme over S , is well reflected in $X(S')$ —better than in the underlying set of X . In the case of Example 1.2, this underlying

set is not a group, whereas every $\mathrm{GL}_N(S')$ is one. More precisely, it is useful to attach to the scheme X over S the contravariant functor from S -schemes to sets

$$S' \longmapsto \text{the set } X(S') \text{ of } S'\text{-points of } X.$$

In the category of S -schemes, this is simply the representable functor

$$h_X : S' \longmapsto \mathrm{Hom}(S', X)$$

attached to X . By Yoneda's lemma, the functor $X \mapsto h_X$ is fully faithful. Rather than thinking of an S -scheme X as a ringed space, endowed with $X \rightarrow S$ and satisfying suitable properties, it is often convenient to think of it as a functor

$$S\text{-points} : (\mathrm{Schemes}/S)^0 \longrightarrow (\mathrm{Sets}),$$

with the virtue of being representable. When one wants to define a “moduli space,” the first step is to define the corresponding functor. Typically, defining this functor requires a relative theory.

Example 3. Let X be projective over a field k . Question: what does “the moduli space of closed subschemes of X ” mean? For $u : S' \rightarrow \mathrm{Spec}(k)$, let X' over S' be deduced from X by base change. Let $\mathcal{H}(S')$ be the set of closed subschemes Z' of X' , flat over S' —flat and of finite presentation if one does not want to assume S' noetherian. Answer: it is a scheme $\mathrm{Hilb}(X)$ representing the functor $S' \mapsto \mathcal{H}(S')$.

To be viable, this point of view requires methods for verifying when a functor is representable. Most of the lectures of FGA are devoted to this problem. A definitive solution, extending those works, was obtained by M. Artin [1969], at least if one agrees to replace the category of schemes by the more natural category of algebraic spaces. More natural: in the same sense in which the étale topology is more natural than the Zariski topology.

2 Topoi

Let $u : S' \rightarrow S$ be a morphism of schemes, the “base-change morphism.” Descent theory [Grothendieck FGA, Séminaire Bourbaki 190, 1959–60] considers problems of the following types. *Descent of properties:* let X be a scheme over S , and let X' over S' be deduced from X by base change. Suppose that X'/S' has a property P . Can one conclude that X/S has P ? *Descent of morphisms:* let X, Y be over S , let X', Y' be deduced by base change, and let $g' : X' \rightarrow Y'$ be a morphism of S' -schemes. When does g' come by base change from a morphism $g : X \rightarrow Y$? *Descent of objects:* let X' be over S' . What descent datum on S' is required to construct X over S from which X' is obtained by base change?

If S' is the disjoint union of the opens $(U_i)_{i \in I}$ of a covering of S , base change to S' is essentially restriction to each U_i , and the preceding problems are problems of localization on S and of gluing. Gluing typically requires considering the pairwise intersections $U_i \cap U_j$, and in terms of $u : S' \rightarrow S$, the disjoint union of the $U_i \cap U_j$ is simply the fiber product $S' \times_S S'$.

The theory of topoi makes it possible to transpose topological intuition into descent theory. For $u : S' \rightarrow S$, a base-change morphism of a kind considered in descent theory, for instance faithfully flat and quasi-compact (fqc), base change from S to S' becomes a localization. A descent datum is the analogue of a gluing datum.

An antecedent of descent theory is Galois descent, corresponding to S the spectrum of a field k and S' the spectrum of a Galois extension. Here $S' \times_S S'$ is a sum of copies of S' indexed by $\mathrm{Gal}(k'/k)$. The proofs, and in particular Galois theory, are nevertheless simpler in the more general framework of descent theory. In Cartier's phrase: Grothendieck proves Galois theory, and Galois descent, by Galois descent.

The tool provided by the theory of topoi made possible the construction of étale cohomology of schemes, and this is its best-known success. A sheaf on X for the Zariski topology is a contravariant functor from the category of Zariski opens of X to sets, satisfying a gluing condition for $(U_i)_{i \in I}$, a Zariski open cover of U . In the étale topology, the Zariski site considered above is replaced by the étale site: the category of Zariski opens is replaced by the category of maps $f : U \rightarrow X$ étale over X , and coverings are replaced by covering families $(U_i)_{i \in I}$, i.e. a surjective morphism of schemes over X from the disjoint union of the U_i to U . An antecedent was Serre's introduction of the notion of an isotrivial principal homogeneous space, namely one locally trivial for a topology close to the étale topology. In his Kansas [1955] and Tohoku [1957] articles, Grothendieck had shown that, once a category of sheaves is given, a notion of cohomology groups follows from it. Thus the étale topology provides étale cohomology. That the resulting H^1 groups are reasonable is not surprising. The miracle is that the higher H^i are reasonable as well.

For Grothendieck, the importance of topos theory goes far beyond the sole case of the étale topology. The title given to SGA 4 bears witness to this: "Theory of topoi and étale cohomology of schemes."

Other applications:

(A) *Crystalline cohomology.* Crystalline cohomology is the cohomology of the crystalline topos, and this definition makes its functoriality clear. The crystalline topos is nevertheless delicate to use, and it is often necessary to pass to an interpretation in terms of de Rham complexes.

(B) *Rigid analytic spaces.* Tate's rigid analytic sheaves are coherent sheaves on a suitable ringed topos, and their cohomology is the corresponding cohomology.

(C) *Foliations.* A variety X endowed with a foliation $\mathcal{F}$ defines a "quotient" $X/\mathcal{F}$, which is a topos locally isomorphic to the topos of sheaves on a variety. This viewpoint seems to have been eclipsed by Connes's, which instead associates to $\mathcal{F}$ a noncommutative C^* -algebra.

I would also like to mention Grothendieck's use of "large" sites, or topoi, especially to interpret classifying spaces.

3 The six operations

The formalism of the six operations presupposes the formalism of derived categories. For the history of the latter, I refer to Illusie's text [1990]. The basic idea is this: for all sorts of cohomology groups, their definition supplies not only these groups but also a complex K whose cohomology groups they are. Typically, this complex is not uniquely determined, but it is determined up to quasi-isomorphism: for two variants K' and K'' of K , there is a K''' and morphisms $K' \leftarrow K''' \rightarrow K''$ inducing isomorphisms on cohomology. In the derived category, K becomes unique up to unique isomorphism.

The six operations are functors between derived categories. They are the derived tensor product $\overset{L}{\otimes}$, the internal Hom functor RHom , which gives rise to the local $\mathcal{E}xt^i$, and, for a morphism of schemes $f : X \rightarrow Y$, two direct-image functors Rf_* and $Rf_!$, and two inverse-image functors Lf^* and $Rf^!$.

The formalism of these operations was first isolated in the cohomology of coherent sheaves [Grothendieck 1963]. Here, for direct images, one should consider only proper morphisms, for which $Rf_* = Rf_!$. The formalism gives a relative version of Serre duality, in the form of an adjunction between the functors $Rf_!$ and $Rf^!$. It also gives a fixed-point formula more general than the contemporaneous "Woods Hole fixed point formula" (Summer Institute on Algebraic Geometry, Whitney estate, Woods Hole, Massachusetts, 1964).

Miraculously, the same formalism applies in étale cohomology, with very different proofs. It again provides a duality formalism, of Poincaré type, and Lefschetz-type fixed-point formulae.

4 Motives

Let X be an algebraic variety over an algebraically closed field k . For each prime number ℓ prime to the characteristic, the étale topology provides ℓ -adic cohomology groups $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$. If k is a subfield of $\mathbb{C}$, one has comparison isomorphisms

$$H^i(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell \xrightarrow{\sim} H_{\text{ét}}^i(X, \mathbb{Z}_\ell).$$

For k of characteristic > 0 , there is no functorial integral cohomology giving rise to such isomorphisms. Nevertheless, the $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$, as ℓ varies, have a “family resemblance.” For $i = 1$, and X projective and smooth, $\text{Pic}^0(X)$ plays the role of the nonexistent integral theory: it recovers the $H_{\text{ét}}^1(X, \mathbb{Z}_\ell)$ and is an object “over $\mathbb{Z}$,” in the sense that the groups of homomorphisms between abelian schemes are finitely generated.

The theory of motives is first of all an attempt to find a substitute for the nonexistent integral cohomology, explaining the family resemblance among the $H_{\text{ét}}^i(X, \mathbb{Z}_\ell)$, especially for X projective and smooth. One recognizes the Master’s touch in the idea that the problem is not to define what a motive is: the problem is to define the category of motives and to isolate the structures it carries. These structures should allow one to prove the Weil conjecture on the model of Serre [1960]. See Grothendieck [1969].

It was on this occasion that Grothendieck invented the notion of a Tannakian category, developed in Saavedra’s thesis [1972]. The point was to formalize the tensor product of motives, corresponding by the Künneth formula to the product of varieties.

Even though the theory of motives has not achieved its original goal, its influence has been great. I refer to the Seattle conference on motives [Jannsen et al., 1994] for a panorama of its applications.

Bibliography

A bibliography of Grothendieck is given at the beginning of his *Festschrift* [Cartier et al. 1990, p. xiii–xx].

Other sources: the introduction, and J. Dieudonné’s article “From functional analysis to the foundations of algebraic geometry,” in the same *Festschrift* (p. 1–14); the lectures at the International Congresses of Mathematicians by J.-P. Serre (Stockholm 1962), “Algebraic geometry” (p. 190–496), J. Dieudonné (Moscow 1966), “The work of Alexander Grothendieck” (p. 21–24), and, for Riemann–Roch and Grothendieck groups, that of H. Cartan (Moscow 1966), “The work of Michael F. Atiyah” (p. 9–14).

ARTIN (M.)

[1969] Algebraization of formal moduli, I, in *Global Analysis* (Papers in honor of K. Kodaira), Tokyo: Univ. Tokyo Press, 1969, p. 21–71.

CARTIER (P.), ILLUSIE (L.), KATZ (N.M.) ET AL., eds.

[1990] *The Grothendieck Festschrift*, vol. I, vol. 86 of Progress in Mathematics, Boston: Birkhauser, 1990.

GROTHENDIECK (A.)

[1955] A general theory of fibre spaces with structure sheaf, University of Kansas, 1955.

[1957] Sur quelques points d’algèbre homologique, *Tohoku Math. J.*, **9** (1957), p. 119–221.

[1963] Residues and duality, prenotes for a “Hartshorne seminar,” manuscript. See Hartshorne [1966].

[1969] Standard conjectures on algebraic cycles, in *Algebraic Geometry* (Colloquium Tata Inst., Bombay, 1968), Oxford Univ. Press, 1969, p. 193–199.

[1985] *Recoltes et semailles: Reflexions et temoignages sur un passe de mathemate-cien*, Montpellier: Univ. Sci. et Tech. Languedoc et CNRS, 1985.

[EGA] *Elements de Geometrie algebrique*. Inst. Hautes Etudes Sci. Publ. Math., 4, 8, 11, 17, 20, 24, 28, 32. In collaboration with J. Dieudonne.

[FGA] *Fondements de la Geometrie algebrique*. Extracts from Seminaire Bourbaki 1957–1962: seminars 149, 182, 190, 195, 212, 221, 232, and 236.

[SGA] Seminaire de Geometrie algebrique du Bois Marie. The seminar SGA 5: Cohomologie ℓ -adique et Fonctions L (1965–66), was finally published as Lecture Notes in Math. 589, Berlin-Heidelberg: Springer, 1977.

HARTSHORNE (R.)

[1966] *Residues and duality*, Lecture Notes in Math. 20, Berlin-Heidelberg: Springer, 1966.

ILLUSIE (L.)

[1990] Categories derivees et dualite, travaux de J.-L. Verdier, *Enseign. Math.*, **36** (1990), p. 369–391.

JANNSEN (U.), KLEIMAN (S.) AND SERRE (J.-P.), eds.

[1994] *Motives*, Proceedings of Symposia in Pure Mathematics 55, Providence: Amer. Math. Soc., 1994.

SAAVEDRA (N.)

[1972] *Categories tannakiennes*, Lecture Notes in Math. 265, Berlin-Heidelberg: Springer, 1972.

SERRE (J.-P.)

[1960] Analogues kahleriens de certaines conjectures de Weil, *Ann. of Math.*, **71** (1960), p. 392–394; Œuvres, vol. II, Berlin-Heidelberg: Springer, 1986, p. 1–3.